Position Dilution of Precision (PDOP) is the DOP value used most commonly in GPS to determine the quality of a receiver's position.

It is usually up to the GPS receiver to pick satellites which provide the best position triangulation.

Some GPS receivers allow DOP to be manipulated by the user.

Another reason that error occurs is atmospheric effects, the most common being due to the Earth's ionosphere. The ionosphere slows down and deflects electromagnetic signals passing through it. This error is least when the satellite is directly overhead and is greater when it is nearer the horizon, because then the signal has a greater distance to cover through the atmosphere.

Another kind of error is the multipath error. These are due to the reflective properties of radio waves. Sometime the satellite radio waves bounce off objects and are reflected before reaching the receiver. This causes a delay and thus a slight error. However, nowadays, some techniques are deployed to control both of these kinds of errors.

GPS has also recently taken on a new role. This takes the form of satellite navigation in cars and other places and can help a person plan routes and trips in advance. Knowing where the device is in space is one thing, but it is fairly useless information without something to compare it with. Thus, the mapping part of any GPS software is very important; this is how GPS systems works out possible routes, and allows the user to plan trips in advance, and directs the user on the way.

The GPS receiver selects three satellites depending on signal strength to calculate its position

The maps increase the prices of GPS systems. They are loaded into the system and need to be updated quite often. There are, however, several kinds of map, and each is intended for different users, with different needs.

A road user who is planning a trip will only require accurate information about the roads in the region where he is headed. He will not require much information about the terrain of the land, altitudes and other things that a trekker will perhaps require.